

Compositional Verification of Robust Mixture-of-Experts Architectures

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Abstract

Mixture-of-Experts (MoE) architectures offer modularity and scalability, yet the interaction between routing stability and expert robustness in heterogeneous MoEs remains underexplored. Verifying full MoE systems is intractable: dynamic routing induces combinatorial behavior, and model scale overwhelms current verifier capacity. We present a **compositional verification framework** that decomposes this problem: if the router provably maintains consistent expert selection within an adversarial perturbation region, and the selected expert is provably robust, then the entire MoE is provably robust—reducing verification from the full system to independent components (20× parameter reduction).

Empirically, we show that expert robustness dominates system behavior (13.1 percentage point improvement with adversarial training) while router training has negligible impact (<0.1%) for dissimilar domains. Verified routers achieve 100% certified accuracy at practical threat models. These results establish a principled path toward scalable, formally verified modular AI.

1 Introduction

Formal verification of neural networks provides provable robustness guarantees critical for safety-critical deployment. However, verifying Mixture-of-Experts (MoE) systems poses unique challenges: dynamic routing creates input-dependent computation paths, and the scale of standard MoEs exceeds the capacity of current complete verifiers like α - β -CROWN. To make verification tractable, we treat routing and expert prediction as separable subproblems and show that their guarantees compose cleanly.

We address this challenge through a **compositional verification** approach. Our central result is:

Compositional Robustness Theorem: *For top-1 routing MoE, if the router maintains invariant expert selection within an ℓ_∞ perturbation ball, and the selected expert is robust within that ball, then the MoE system is robust.*

This decomposition is the key insight: rather than verifying the intractable full system, we verify components independently and compose guarantees. The practical impact is a **20× reduction** in verification complexity (2M $\rightarrow$ 96K parameters in our experiments).

A *heterogeneous MoE* combines domain-specific experts (potentially with different architectures or training data) coordinated by a learned router. Unlike homogeneous MoEs where experts share structure, heterogeneous systems require the router to perform domain identification—a distinct task from within-domain classification.

Contributions.

1. **Compositional Verification Framework:** We formalize conditions under which MoE robustness decomposes into router stability and expert robustness, enabling **20× verification speedup** and incremental expert integration without re-verifying the full system.
2. **Expert Dominance Finding:** Expert training determines system robustness (13.1% adversarial accuracy difference); router adversarial training has negligible impact (<0.1%) for visually dissimilar domains, enabling 4× training speedup.

3. **Verification-Aware Design:** We identify architectural requirements for tractable MoE verification and demonstrate that adversarial training is a *prerequisite* for formal verifiability.

2 Background and Related Work

Mixture-of-Experts. The MoE paradigm [1] decomposes problems into specialized subtasks via gating networks. Shazeer et al. [2] scaled this through sparse top- k routing, and Riquelme et al. [3] extended MoE to vision. Recent work explores heterogeneous compositions [4] and modular architectures [5], but formal robustness properties remain unaddressed.

Neural Network Verification. Complete verifiers like α - β -CROWN [6] provide sound robustness certificates via bound propagation and branch-and-bound. Certified training methods (CROWN-IBP [7], SABR [8]) improve verifiability by training with formal bounds. Randomized smoothing [9] offers probabilistic certificates for larger models. However, all approaches face scalability limits with multi-million parameter systems.

Compositional Verification. Compositional approaches decompose verification into manageable sub-problems. In robotics and hybrid systems, assume-guarantee reasoning enables modular certificates [10]. For neural networks, modular verification remains nascent. MoEs are uniquely suited to compositional verification because the router explicitly partitions inputs into disjoint expert domains—if this partitioning is stable under perturbation, component guarantees compose directly.

Research Gap. Prior work addresses homogeneous MoE scalability or monolithic neural network verification. We provide the first compositional verification framework for heterogeneous MoE systems, bridging modular architecture design with formal robustness guarantees.

3 Compositional Verification Framework

3.1 Problem Setup

Let $x \in \mathbb{R}^d$ be an input with label y . A heterogeneous MoE consists of N domain-specific experts $\{E_i : \mathbb{R}^d \rightarrow \mathbb{R}^{C_i}\}_{i=1}^N$ and a router $\mathcal{G} : \mathbb{R}^d \rightarrow \mathbb{R}^N$ selecting expert $m^* = \arg \max_i \mathcal{G}_i(x)$. We consider ℓ_∞ -bounded perturbations $B_\varepsilon(x) = \{x' : \|x' - x\|_\infty \leq \varepsilon\}$.

3.2 Main Result

We now formalize the compositional robustness guarantee.

Theorem 1 (Compositional Robustness). *Let $\text{MoE} = (\mathcal{G}, \{E_i\})$ use top-1 routing and let $m^* = \arg \max_i \mathcal{G}_i(x)$. If*

$$\forall x' \in B_\varepsilon(x) : \arg \max_i \mathcal{G}_i(x') = m^* \quad (\text{router stability})$$

and

$$\forall x' \in B_\varepsilon(x) : \arg \max E_{m^*}(x') = y \quad (\text{expert robustness}),$$

then the MoE is ε -robust at x .

Intuition. The router partitions the input space into disjoint expert regions; if these partitions are stable under perturbation and the selected expert is robust within its region, the entire system inherits robustness.

Proof. For any $x' \in B_\varepsilon(x)$: router stability guarantees $\arg \max_i \mathcal{G}_i(x') = m^*$, so expert E_{m^*} processes x' . Expert robustness guarantees $\arg \max E_{m^*}(x') = y$. Since MoE output depends solely on the selected expert's prediction, the system output remains y for all $x' \in B_\varepsilon(x)$. $\square$

Remark 1 (Top-1 Requirement). *Theorem 1 requires top-1 routing. For top- k routing ($k > 1$), the weighted combination of multiple experts introduces additional attack surface: adversaries could exploit the weighting mechanism even if individual experts are robust. Extending compositional guarantees to top- k requires verifying both expert selection stability and weight stability—a stronger condition we leave to future work.*

Practical Impact. This decomposition transforms an intractable problem (verifying 2M+ parameter MoE) into manageable subproblems (verifying 96K parameter components). Verification cost scales with the *largest component*, not the full system. New experts integrate without re-verifying existing components.

3.3 Verification Implementation

We use α - β -CROWN for component verification:

Router Verification: For input x with selection m^* , verify $\mathcal{G}_{m^*}(x') - \mathcal{G}_j(x') > 0$ for all $j \neq m^*$ and $x' \in B_\varepsilon(x)$. This reduces to $N - 1$ binary properties per input.

Expert Verification: Standard robustness verification—verify the true class logit exceeds all others within $B_\varepsilon(x)$.

Challenges: Router verification difficulty depends on decision margin geometry. When domains are visually dissimilar (e.g., CIFAR-10 vs MNIST), routers learn large margins ($\gg \varepsilon$), enabling easy certification. Similar domains produce smaller margins and harder verification (see Section 5).

3.4 Verification-Aware Architecture

Both router and experts use **VerifiableCNN** (96K parameters) designed for α - β -CROWN: average pooling (linear, unlike max pooling which introduces additional branching), no BatchNorm (folded into weights), minimal operators. Training uses PGD adversarial training ($\varepsilon = 8/255$) for robust variants. Full specifications in Appendix B.

4 Experiments

We evaluate on CIFAR-10 and MNIST—visually dissimilar domains enabling controlled analysis of compositional effects. Extended experiments on similar domains appear in Appendix C.

4.1 Router Verification Enables Compositional Guarantees

Training	Certified Routing Accuracy (%)		
	$\varepsilon = 2/255$	$\varepsilon = 4/255$	$\varepsilon = 8/255$
Non-Robust (NRT)	100	100	60
Robust (RT)	100	100	100

Table 1: Router verification (40 samples, 5 runs). Both achieve 100% at practical threat models ($\varepsilon \leq 4/255$). No falsifications—all failures are timeouts. RT extends certification to larger perturbations.

Key Finding: Routers achieve high formal certification even without adversarial training at practical ε . The router achieves high certified accuracy because CIFAR-10 and MNIST lie on well-separated manifolds, yielding large decision margins far exceeding ε . The router’s empirical adversarial accuracy ($\sim 35\%$) differs from its (100%) certified accuracy because heuristic attacks explore limited local gradient directions, while certification reasons about all perturbations within $B_\varepsilon(x)$. This reflects the distinction between heuristic attack success and worst-case guarantees: PGD finds *some* adversarial examples, while formal verification proves *none exist* that change routing.

Expert	Training	Certified Robustness Accuracy (%)		
		$\varepsilon = 2/255$	$\varepsilon = 4/255$	$\varepsilon = 8/255$
CIFAR-10	NRT	0	0	0
	RT	90	54	0
MNIST	NRT	75	0	0
	RT	100	99	95

Table 2: Expert verification (20 samples each, 5 runs). Robust training enables verification: CIFAR-10-NRT is completely unverifiable; CIFAR-10-RT achieves 90% at $\varepsilon = 2/255$.

4.2 Expert Verification Requires Robust Training

Key Finding: Robust training is a *prerequisite* for verifiability, not merely beneficial. Non-robust CIFAR-10 experts produce highly irregular decision boundaries, which lead to loose CROWN bounds and universal verification failure. Adversarial training smooths boundaries, enabling tractable formal analysis.

4.3 Expert Robustness Dominates System Behavior

Configuration	Clean Acc (%)	Adv Acc (%)
Both Experts RT	87.98	41.93
Both Experts NRT	91.32	28.78
<i>Difference from expert config</i>	<i>3.34</i>	<i>13.15</i>
<i>Difference from router config</i>	<i><0.1</i>	<i><0.1</i>

Table 3: MoE compositional robustness (8 configurations $\times$ 5 runs). Expert configuration causes 13.15% adversarial accuracy difference; router training (RT vs NRT) causes $<0.1\%$ difference across all metrics.

All configurations achieve **100% routing accuracy** (clean and adversarial), validating the compositional property: the router acts as a deterministic switch, accurately transferring expert robustness to the system.

Practical Implication: For dissimilar domains, router adversarial training can be omitted ($4\times$ speedup, 968s vs 4107s) and focus resources on expert robustness.

4.4 Scalability

Compositional verification achieves **$20\times$ parameter reduction** (2M full MoE $\rightarrow$ 96K largest component) and **4.6s average verification time** per sample at practical ε . New experts integrate by verifying only the new component—existing verifications remain valid. Training efficiency improves 16.7% for 3-expert systems via frozen-expert router fine-tuning.

5 Discussion and Limitations

When Does Compositional Verification Apply? The framework requires router stability within $B_\varepsilon(x)$. For dissimilar domains (CIFAR-10 vs MNIST), routers learn large decision margins enabling easy certification. For similar domains (e.g., GTSRB vs PTSD traffic signs), routing accuracy degrades to 93%, and compositional guarantees degrade unless the router achieves sufficiently large margins; this may require stronger adversarial training or probabilistic certification.

Architecture Scope. Our verification-compatible CNNs (96K parameters) demonstrate the framework but limit expressiveness. Scaling to Transformers faces verifier memory constraints. Promising directions include verification-aware architecture search and combining compositional decomposition with randomized smoothing for probabilistic certificates on larger models.

Top- k Extension. Theorem 1 applies to top-1 routing. Top- k ($k > 1$) requires verifying weight stability in addition to selection stability—a harder condition requiring future work.

Relationship to Empirical Robustness. Adversarial training improves certifiability by smoothing decision boundaries, reducing bound looseness in CROWN analysis. This explains why robust training simultaneously improves empirical adversarial accuracy and formal certification rates.

6 Conclusion

We establish that MoE robustness composes modularly under top-1 routing: verified router stability combined with verified expert robustness yields verified system robustness. This decomposition achieves $20\times$ verification speedup and enables incremental expert integration.

Our experiments reveal that expert robustness dominates system behavior (13.1% impact) while router training is negligible for dissimilar domains ($<0.1\%$). Critically, robust training is prerequisite for formal verifiability—non-robust components are generally incompatible with sound verification at practical scales.

These results demonstrate that compositional design enables tractable formal verification of modular AI systems, advancing the path toward verifiable, safety-critical deployment.

Future work includes extending compositional guarantees to top- k routing and larger transformer-based MoE systems.

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A Training Configuration

Experts: Adam, LR 10^{-3} , 200 epochs, batch 128. RT: PGD ($\varepsilon = 8/255$, 7 steps, step 2/255).

Router: Adam, LR 10^{-3} , 50 epochs on frozen experts. Same PGD for RT.

B Architecture

VerifiableCNN (96K params): 4 conv blocks ($3 \rightarrow 20 \rightarrow 28 \rightarrow 40 \rightarrow 56$ channels), 3×3 kernels, ReLU, AvgPool. Global average pooling $\rightarrow$ FC(56, 128) $\rightarrow$ ReLU $\rightarrow$ FC(128, C). No BatchNorm (folded for ONNX).

C Similar Domain Experiments

GTSRB + PTSD (both traffic signs): routing accuracy degrades to 93.17% (vs 100% for CIFAR-10 + MNIST), adversarial accuracy drops to 29.47% (vs 41.59%). The 6.83% routing error propagates to 12.12% system accuracy reduction.

D Hardware

Intel i7-14700K, RTX 4060 8GB, 64GB RAM. PyTorch 2.1, α - β -CROWN (Dec 2024), Gurobi 11.0. Timeout: 300s/sample.